Revisiting Elasticities of alcoholic and non-alcoholic Beverages in Colombia

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Abstract

Unhealthy diets cause Non-Communicable Diseases, and over-consumption of calorie-dense food, like Sugar-Sweetened Beverages (SSB), is a major driver of unhealthy diets. Therefore, taxes on SSB have been proposed as an effective policy for primary prevention, but accurate estimates of the price and income elasticities of SB are crucial for the design of those taxes.This paper identifies and corrects biases the in estimation of price and income elasticities of Sweetened Beverages (SB) in Colombia. We use data from Colombia's 2016-2017 Household Survey of Income and Expenditures (ENPH) to estimate a Deaton's Unit Value Model (DUVM) for SB and potential substitutes. Our methods adjust for biases from measurement error from uncoded consumption, censoring from zero consumption, and exclusion of rural households; to derive own-price, cross-price, and income elasticities on-trade and full market (including consumption away from home). Censoring for zero consumption, exclusion of rural households, and omission of uncoded consumption were found to be the most important sources of bias of elasticities of SB. When adjusted for all biases, the price elasticity of SB is estimated at -0.50, and the biases can lead to estimates of -0.36. Also, it is not possible in Colombia to get a separate estimate for Sugar-Sweetened Beverages and Non-Sugar-Sweetened Beverages.Substitution towards alcoholic beverages is a public health concern supported by our findings. Policy design must be cautious about using unbiased estimates of elasticities to have an accurate design of the tax.

Keywords: Sweetened Beverages, elasticities, Colombia.

JEL: D12, I12, C13

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1 Introduction

Non-Communicable Diseases (NCDs) concentrate most of the burden of disease worldwide [1] and represent a threat to global health [2] and, more generally, to progress on sustainable development [3, 4, 5]. Unhealthy diets are one of the five common risk factors of NCDs [6, 7]. Consumption of Sugar-Sweetened Beverages (SSB) play an important role in unhealthy diets [8, 9], and even Non-Sugar-Sweetened Beverages (NSSB) might have some effects on health [10, 11, 12], making a case for health policy to reduce consumption of Sweetened Beverages (SB) [13]. Furthermore, from the wider perspective of sustainable development, soft drinks are not part of a healthy diet that supports environmental sustainability [14].

Consumption of is one component of households' expenditure on food, and expenditure decisions are strongly dependent on prices and income. Such dependence makes fiscal policies, in particular excise taxes on the consumption of SB, a cost-effective intervention on unhealthy diets: a higher excise tax increases SB prices, making them less affordable and changing beverage consumption decisions [15, 16]. For that reason, excise taxes have become one of the recommended interventions to control the epidemic of NCDs [17, 18, 19] and a policy instrument for sustainable development [14]. In order to accurately design an excise tax on SB, it is pivotal to have a good understanding of both the individuals' and households' consumption and the responsiveness of that consumption to changes in prices; any bias in those elements is undesirable from a policy perspective, as it directly translates into overestimation or underestimation of the expected health benefits and tax revenues, misguiding policy design and advocacy.

In the case of SB, estimation of consumption and elasticities faces several potential sources of bias, that are frequently overseen. First of all, excise taxes on SB are a public health intervention as they intend to work as an effective intervention for the whole population [20]; therefore, evidence at the individual, clinical or community levels from case studies or randomized control trials, although informative, has a limited scope to inform policy design at the population level, given the limitation they have on external validity [21]. This problem calls for using population-representative data such

as household surveys from national statistical systems, especially expenditure surveys that have data on both the consumption decision and on the main economic drivers (income and price). The measurement error in those surveys is another source of bias [22]; particularly relevant is the measurement error in consumption of SB because surveys with economic variables register data on the consumption of SB instead of intake and also because consumption combines decisions at both the individual and the household level. Other biases might emerge from overlooking the particularities of the market of SB. One of them is product reformulation, meaning that the soda industry has gradually introduced a wide portfolio of new products with sweeteners other than sugar and products with significantly reduced levels of sugar or calories [23]; the implication in biases is that such diversity of products makes it more difficult to separate consumption of products with added sugar from the ones with no added sugar in household surveys. Another complexity of the market is the set of potential substitutes. Even though, in principle, they are limited to beverages, in practice there is wide diversity in the categories of beverages and in the products in each category. Furthermore, consumption patterns among categories change broadly worldwide, making it hard to extrapolate estimates from one country or region to another one. Finally, regarding the estimates of elasticities, there are biases coming from using reduced forms instead of structural forms for estimation, the simultaneous modeling of SB and potential substitutes, and the proper modeling of zero consumption.

The purpose of this article is to identify and correct biases in the estimation of price and income elasticities of SB in Colombia. We do so by estimating price and income elasticities in a demand system for beverages, using a dataset that is adjusted to correct the bias of unfeasible consumption. The ultimate goal of the paper is to reduce misleading biases in the technical discussion of taxes on SB in Colombia, and in that way, to help in constructing a more precise and technical discussion about the policy and its expected effects.

2 Materials and methods

2.1 Model

We estimate price elasticities of demand using Deaton's Unit Value Model (DUVM) [24], a commonly used method for estimating the price elasticity of demand for goods and services, particularly in developing countries where detailed market data may not be available. The model is based on the assumption that consumers have a fixed budget for spending on a particular good and therefore adjust their consumption levels in response to changes in price. Estimating a demand system requires information on household expenditures at different price levels. The price can vary spatially. However, accurate price data for different goods and their regional variation is not readily available in Colombia. This scarcity of data justifies switching to the unit value model. In particular, for the case of large consumption groups with weak separability, we use DUVM to estimate the price and income elasticities, using unit values, and where the model enables us to compensate for the quality effect.

For the estimation of price and income elasticities, We follow the steps proposed by Deaton. First, the unit values are derived from the ENPH at the household level. This is done by dividing reported total expenditure on the particular product by their corresponding quantity $\nu_{ghc} = \frac{x_{ghc}}{q_{ghc}}$. where $\nu_g hc$, $x_g hc$ and $q_g hc$ are respectively the unit value, expenditure, and quantity of good g in household h located in cluster c. Second, we test whether obtained unit values in step 1 satisfy the main identifying assumption: unit values vary spatially. Third, Deaton proposes estimating unit value and budget share equations:

$$ln\nu_{ghc} = \alpha_g^1 + \beta_g^1 lnx_{hc} + \gamma_g^1 z_{hc} + \sum_{j=1}^{N} \Psi_g lnp_{jc} + u_{ghc}^1 \tag{1}$$

$$w_{ghc} = \alpha_g^0 + \beta_g^0 lnx_{hc} + \gamma_g^0 z_{hc} + \sum_{j=1}^{N} \theta_g lnp_{jc} + (f_{gc} + u_{ghc}^0) \tag{2}$$

Where w_{ghc}, in Eq 2, is the budget share for household h, located in area c, allocated to acquiring

good g. xhc is the total household expenditure, pjc is the price of the N goods that the household can purchase and this is assumed not to vary within area c. z_hc is a vector of household characteristics and f is an error associated with the area (cluster), while u is an idiosyncratic error (i.e., for each household). Therefore, the budget shares each household allocates for each good depend on the household's total budget, the inherent characteristics of the household, and the price of all goods purchased by each household. For Eq 1, ν_{ghc} is the unit value of good g for household h (in area c) and is a function of the same variables that affect w_{ghc} (except for the cluster level error).

The fourth step involves stripping the household level demand and unit values of the effects of household expenditure and household characteristics and then averaging across clusters since the primary interest is to estimate elasticity at the cluster level (Eq 3).

$$\tilde{w}_{gc} = \alpha_g^2 + \theta_g^2 ln(\tilde{\nu}_{cg}) + u_{gc}^2 \tag{3}$$

At the fifth and final stage, the effect of price on the budget share is ex-tracted from the ratio by use of the theory linking quality and quantity elasticities. Price elasticity for good g $e_{gj} = \frac{\theta_g}{w_g} - \delta_{gj}$, where $\delta_{gj} = I(g = j)$ Kronecker's delta. For more details, see Deaton [24]

2.2 Data and estimation

For estimation of elasticities, we use the National Household Budget Survey 2016-2017 (ENPH), which is the most recent income and expenditure survey in Colombia; the survey is administered by the National Department of Statistics (DANE). The survey collects data on households' amount and sources of income and on the amount and distribution of expenditure. The ENPH has a probabilistic, stratified, three-stage sample design, with a sample of 87,201 households (HH) with 291,590 individuals (II), designed to be statistically representative of the 14,350,388 households and 48,018,203 people in Colombia in 2016. . The survey is accurate for the analysis because it has data on food consumption decisions and expenditure at both the individual and the household levels.

Other national surveys are not accurate because they do not have economic variables such as income or prices, as is the case with the National Survey on Nutrition (ENSIN) is more precise in measuring consumption as it records food intake at the individual level; however, it does not have economic variables such as income or prices. Along the same line, the National Survey of Quality of Life (ECV) allows the identification of households that consumed SB, but it does not have data on prices or unit values and potential substitutes.

Estimation proceeds in four steps: (i) identification of consumption, (ii) econometric estimation, (iii) calculation of elasticities, and (iv) identification of biases. For the first step, we identify all coded and uncoded consumption of SB and potential substitutes in the survey, to correct for the bias of underestimation of consumption, following the method proposed by [25]. This correction is important because nationally representative surveys have reported 67% of individuals consuming SSB (Encuesta de Calidad de Vida) and a prevalence of intake of soda drinks of 83.3% [26], which has a huge difference with previous estimates of elasticities based on a prevalence of 32% [27]. Coded consumption follows from identification of beverages in the Classification of Individual Consumption According to Purpose - COICOP [28], while uncoded consumption was identified using regular expression analytics for terms related to beverages (e.g. gaseosa - soda) in records of consumption in string text. All the records of consumption of beverages were classified into one of the following categories: Sweetened Beverages (SB), Beer, Spirits, Water, and Dairy.

For the second step, we estimate the DUVM grouping the households in 354 clusters. Our estimations have the following advantages over previous estimations that arise with some econometric adjustments that are relevant for the general case of nutrition [29, Ch.6] and for the specific case of SB. First, demand responds not only to prices and income but also to other variables representing population heterogeneity, and are usually called in the literature as demographics [30]. We included as control variables the sex, age, and level of education (no education, elementary, high school, college) of the head of the household, household size, and a dummy variable representing whether the household has children or not. Second, the estimation considers the censoring issue of zero consumption caused by the decision not to consume beverages. Otherwise, it will erroneously lead to

assuming corner solutions for households with positive uncoded consumption (e.g. [27]), magnifying the effect of the bias from measurement error. Zero consumption was addressed following a selection bias correction approach. For each good, the routine estimates the beforehand Inverse Mills Ratio (IMR) based on the binary model (Consumption is not nil), and then, it uses the IMR variable in models of the first equation [31].

A pair of additional sources of bias are worth to explore. One is the exclusion of rural households from the analysis, an exclusion that has been used in previous estimates [27]. The other one is the combination of biases clearly identified in previous estimates; in particular, we have identified that estimates in [27] only include urban households and do not adjust for uncoded consumption.

The third step in the estimation was the calculation of elasticities. Considering that reactions to price changes at off-trade (consumed at home) and on-trade (consumed away from home these two might be different, we estimate the elasticities for the off-trade channel and the full market (off-trade plus on-trade). Finally, for the fourth step, we estimated the biases by comparing our estimators with the ones obtained from missing some of the adjustments mentioned before. All estimations were carried out in Stata 17.0, and the demand system estimated was estimated using the WELCOM tool version 2.1 [31]. Standard errors for all elasticities were obtained by bootstrapping the sample 100 times.

3 Results

Households in Colombia tend to have a male household head (63%) and on average 3.37 members per household. Households in majority are urban (82%). (Table 1).

Households' expenditure on SB is higher than for the other drinks, except dairy drinks. This finding holds if we consider just codified consumption or if it is corrected for uncodified consumption. Mean monthly soft drink consumption among households in the off-trade market is 3.99 liters, while in the full-market, it is 8.4 liters on average per month (Table 2).

Elasticity estimates are shown in Appendix B. Tables 3 and 6 present the price and income elasticities for the categories of beverages. The first column (model 1) presents the model with all the adjustments mentioned in section 2, our base estimation, and each of the subsequent columns shows the estimates obtained from dropping one adjustment at a time, with the purpose to identify the bias in price elasticities.

The price elasticity for SB in the full market is estimated at -0.50, meaning that a 10% increase in the price of SB is expected to decrease consumption in 5.0%. Meanwhile, off-trade SB elasticity is -0.35 suggesting that SB consumption for home is less price - sensitive that out-home consumption. Regarding potential sources of biases, exclusion of uncoded consumption (model 5), not weighting (model 3) and exclusion of rural households (model 6) seem to be the most important sources of bias. Also, compared to other categories of beverages, SB seem to be more sensitive than dairy beverages and less sensitive than water (Table 3).

Regarding income elasticities of SB, the estimated elasticity is 0.88 for the full market and is similar off-trade, suggesting that a 10% of increase in income translate in rising SB consumption by 8.8% (Table 6). Our results indicate that all elasticities are greater than zero, which means that considered beverages are normal goods which demand would increase if household income rises. It can also be seen that , although no affect SB income elasticity in high degree, the most important source of bias is not controlling for censoring from zero consumption (model 2), as evidenced in the cases of beer and spirits. Not controlling for nil consumption triple the income elasticity estimates for these goods.

Results for cross price elasticities to examine substitution patterns from the base model are presented in Table 5. The estimates in the first column suggest that all categories of beverages are complementary of SB, that is, an increase in the price of SB triggers a substitution effect across all categories of beverages, but with greater effects off-trade and full market in spirits (-0.55 and -0.62 respectively) and beer (-1.08) full market.

According to our findings, there is variance in SB price elasticity by income group (Table 6); lowest

income households (1st income tercile) have higher own-price elasticities for SB (as well as for beer, water and dairy), meaning that their consumption of SB after a tax would reduce relatively more than in high income households (3rd tercile). This result is robust whether we only include off trade consumption or we analyze the full market.

4 Discussion

The analysis shows that, when controlling for different sources of biases, the price elasticity of SB is -0.50 while the income elasticity is 0.88; it also indicates that those biases can be of considerable magnitude ($\approx$ 0.14 percentage points) and therefore can significantly mislead technical discussions and policy design of taxes on SSB. In addition, SSB cannot be identified because the coding system in the survey (COICOP) cannot differentiate between beverages sweetened with sugar and those sweetened with other substances; furthermore, for uncoded consumption in available surveys such distinction is also impossible.

The estimates presented in the paper have limitations that we have considered and have addressed as far as possible. One important limitation is the difficulty in identifying and accurately measuring the consumption of beverages. The method implemented for identification [25] addressed this issue by doing a complete consideration of all the coded categories and exhaustive use of regular expression (analytics) for uncoded consumption related to beverages. Despite that, it is possible, mostly in uncoded consumption, that some very particular types of beverages were not identified. However, the inclusion of such consumption should play a minor role in estimation. Also, despite the adjustment, there is an important remaining gap between adjusted demand (aggregate consumption) and supply; for the case of SB, reported supply is estimated to be 2.47 times the adjusted demand [25], and such gap is a bias that cannot be corrected with the data in the ENPH survey.

Another major limitation is the impossibility of separating SSB from NSSB. Such distinction is important because the evidence on effects on health is different in history, volume and conclusiveness for each category, and because a differential tax policy for each category should be designed taking

into account the potential substitution between them. Additionally, SB/Alcoholic drinks substitution patterns may differ by the degree of sugar added into SB [32] The systems to classify products do not seem to keep up with product reformulation in SB and substitutes, and that seems to happen not only with the COICOP used in ENPH but also with other classifications such as Harmonized Systems for products in international trade [33] or the Central product Classification (CPC) [34] for product manufacturing; since the problem comes from international classification it might represent a systemic risk in that it might be systematically affecting estimations worldwide. Such lag is inconvenient for the generation of evidence on the consumption of SSB that feeds the design of health policy, as it is not possible to obtain separate estimates for products with high content of harmful substances such as sugar (SSB vs NSSB) or alcohol (e.g. beer vs. Non-Alcoholic beer); also, when this problem is overlooked (e.g.[27] for SSB in Colombia), it becomes an additional source of bias and from there, a source of misguidance for technical discussions in those areas. We addressed that difficulty by doing a careful review of all the coded categories in demand of SB (COICOP) and supply of SB (CPC and Harmonized System) with the purpose to find ways to estimate the consumption for each category separately, either in the observed supply, demand or both. Despite that effort, it was not possible to define a clear distinction between these two categories; this is an interesting result because it suggests that the country needs to overcome institutional barriers in the national system of statistics for keeping track of the production and consumption of both categories, and that technical discussions on taxes on SSB must recognize such structural limitation. Regarding alcoholic drinks, the COICOP classification does not allow us to disentangle on-trade beer consumption, so the elasticities for this good must be interpreted as price elasticity of off-trade beer consumption even in the estimation of the full market elasticities.

An additional caveat for the interpretation of results is related to the consumption of a household as the unit of analysis, which has two implications. First, the estimates cannot be interpreted as the effect on beverage intake, because the survey does not measure intake. Intake is important from the health perspective because it is a more precise measure of exposition to the risk factor at the individual level; however, the survey with information on intake (ENSIN) does not have data on

economic variables to estimate elasticities, and the intake decision is indirectly dependent on prices and income, as opposed to the consumption decision, where prices an income play a direct role. Also, the model does not capture differences between individual consumption and household consumption, as it aggregates both decisions into an aggregate consumption at the individual level. This analysis is relevant because the estimates suggest that there are differences in elasticities from both decisions, as the adjustment for uncoded consumption, which comes mainly from individual consumption, changes significantly the magnitude of elasticities.

The results have important policy implications. First, the biases identified in elasticities have been used in ex-ante evaluation of the implementation of SSB taxes, not only in estimating expected tax revenues [27] but also in estimating the expected effects on health [35]; this, in turn, can significantly mislead the design of the tax and create high expectations in policymakers that might not match the observed results in ex-post evaluations once the tax is implemented. Second, since income plays an important role in the consumption of SB, any tax should consider indexation to income in order to avoid the effect of the tax vanishing over time as real income grows. Third, it is crucial to incorporate in the discussion the differentiation between SSB and NSSB. To our knowledge, the coding systems for consumption and production do not allow for precisely measuring each of these categories, which implies that previous estimates for Colombia are actually elasticities for SB, not for SSB. Since those coding systems are used worldwide, it is possible that such misjudgement is present in estimates of elasticities of SB in other countries.

Finally, taxing SB may reduce their purchase, but assessing the impact on health demands wider consideration of alternative beverage choices. Effects on alcoholic drinks are of particular concern, as alcohol consumption has additional health harms. Changes in consumption of alcoholic drinks may reinforce or negate the intended effect of price changes for soft-drinks [36, 37]. This is the first attempt to measure the interrelationships between SB/ Alcoholic drinks in Colombia. Our results suggest that increasing SB price as a result of taxation, may reduce the consumption of beer and spirits, however increasing the price of spirits may increase the consumption of SB, similar findings have been documented in the international literature [32].

5 Conclusions

Price and income elasticities play a key role in policy design of health taxes. Biases in elasticities can considerably mislead the policy discussion, because they can bias the ex-ante evaluation on expected effects of the implementation of the policy in the areas of health, tax revenue, or any other effect on sustainable development. Censoring for zero consumption, exclusion of rural households and omission of uncoded consumption were found to be the most important sources of bias of elasticities of SB. When adjusted for all biases, the price elasticity of SB is estimated at -0.50, and it is not possible in Colombia to get a separate estimate for SSB and NSSB; this is unfortunate because such separation is crucial for the policy of taxing SSB. Finally, substitution towards alcoholic beverages is a public health concern supported by our findings. Policy design must take such substitution into account because if it does not, the tax might have negative effects on health by increasing the consumption of alcohol.

With regard to policy implications, the estimates presented in this paper suggest that there is space for health taxes on SB in Colombia [38, 39, 15, 16], as a policy instrument on prevention of NCDs: it is an inelastic good with high prevalence of consumption and therefore an important risk factor for unhealthy diets. However, policy design must be cautious on the goods included in the excise tax (tax base), as product reformulation has the potential to change consumption patterns towards new products, and from there, to change the expected effects on health and tax revenues.

References

[1] Murray, Christopher J L et al. “Global burden of 87 risk factors in 204 countries and territories, 1990–2019: a systematic analysis for the Global Burden of Disease Study 2019”. In: *The Lancet* 396.10258 (2020), pp. 1223–1249. DOI: 10.1016/S0140-6736(20)30752-2.

[2] Benziger, Catherine P, Roth, Gregory A, and Moran, Andrew E. *The Global Burden of Disease Study and the Preventable Burden of NCD*. 2016. DOI: 10.1016/j.gheart.2016.10.024.

[3] Engelgau, Michael et al. “The economic effect of noncommunicable diseases on households and nations: a review of existing evidence.” In: *Journal of health communication* 16 Suppl 2 (2011), pp. 75–81. DOI: 10.1080/10810730.2011.601394.

[4] Jaspers, Loes et al. “The global impact of non-communicable diseases on households and impoverishment: a systematic review”. In: *European Journal of Epidemiology* 30.3 (2015), pp. 163–188. DOI: 10.1007/s10654-014-9983-3.

[5] Nugent, Rachel et al. “Investing in non-communicable disease prevention and management to advance the Sustainable Development Goals”. In: *The Lancet* 391.10134 (2018), pp. 2029–2035. DOI: 10.1016/S0140-6736(18)30667-6.

[6] UN. *Political declaration of the third high-level meeting of the General Assembly on the prevention and control of non-communicable diseases (A/73/L.2)*. Tech. rep. United Nations (UN), Resolution adopted by the General Assembly on 10 October 2018, 2018.

[7] PAHO. *NCDs at a Glance: NCD Mortality and Risk Factor Prevalence in the Americas*. Tech. rep. Pan American Health Organization (PAHO), 2019.

[8] Popkin, Barry M and Hawkes, Corinna. “Sweetening of the global diet, particularly beverages: patterns, trends, and policy responses”. In: *The Lancet Diabetes & Endocrinology* 4.2 (2016), pp. 174–186. DOI: https://doi.org/10.1016/S2213-8587(15)00419-2.

[9] Temple, Norman J. and Alp, Kathryn. "Sugar in Beverages: Effects on Human Health". In: *Beverage Impacts on Health and Nutrition*. Ed. by Wilson, Ted and Temple, Norman J. Humana Press, Cham, 2016. Chap. Ch.19, pp. 277–284. ISBN: 978-3-319-23672-8. DOI: 10.1007/978-3-319-23672-8{_}19.

[10] Borges, Maria Carolina et al. "Artificially Sweetened Beverages and the Response to the Global Obesity Crisis". eng. In: *PLoS medicine* 14.1 (Jan. 2017), e1002195–e1002195. DOI: 10.1371/journal.pmed.1002195.

[11] Toews, Ingrid et al. "Association between intake of non-sugar sweeteners and health outcomes: systematic review and meta-analyses of randomised and non-randomised controlled trials and observational studies". In: *BMJ* 364 (Jan. 2019), k4718. DOI: 10.1136/bmj.k4718.

[12] Malik, Vasanti S. "Non-sugar sweeteners and health". In: *BMJ* 364 (Jan. 2019), k5005. DOI: 10.1136/bmj.k5005.

[13] Pomeranz, Jennifer L. "Advanced policy options to regulate sugar-sweetened beverages to support public health". In: *Journal of Public Health Policy* 33.1 (2012), pp. 75–88. DOI: 10.1057/jphp.2011.46.

[14] Willett, Walter et al. "Food in the Anthropocene: the EAT-Lancet Commission on healthy diets from sustainable food systems". In: *The Lancet* 393.10170 (Feb. 2019), pp. 447–492. DOI: 10.1016/S0140-6736(18)31788-4.

[15] Powell, Lisa M and Maciejewski, Matthew L. "Taxes and Sugar-Sweetened Beverages". In: *JAMA* 319.3 (2018), pp. 229–230. DOI: 10.1001/jama.2017.19522.

[16] World Bank. *Taxes on Sugar-Sweetened Beverages*. World Bank, Apr. 2020. DOI: 10.1596/33969.

[17] Sassi, Franco and Belloni, Annalisa. "Fiscal incentives, behavior change and health promotion: what place in the health-in-all-policies toolkit?" In: *Health Promotion International* 29.suppl_1 (June 2014), pp. i103–i112. DOI: 10.1093/heapro/dau050.

[18] Thow, Anne Marie et al. “Fiscal policy to improve diets and prevent noncommunicable diseases: from recommendations to action”. In: *Bulletin of the World Health Organization* 96.3 (2018), pp. 201–210. DOI: 10.2471/BLT.17.195982.

[19] WHO. *Tackling NCDs. ’Best buys’ and other recommended interventions for the prevention and control of noncommunicable diseases*. Tech. rep. Technical Report WHO/NMH/NVI/17.9, World Health Organization (WHO), 2017.

[20] Gillam, Stephen, Yates, Jan, and Badrinath, Padmanabhan. *Essential Public Health. Theory and Practice*. 2nd Ed. Cambridge University Press, 2012. ISBN: 978-1-107-60176-5.

[21] Deaton, Angus and Cartwright, Nancy. “Understanding and misunderstanding randomized controlled trials”. In: *Social Science & Medicine* 210 (2018), pp. 2–21. DOI: https://doi.org/10.1016/j.socscimed.2017.12.005.

[22] Deaton, Angus. *The Analysis of Household Surveys: A Microeconometric Approach to Development Policy*. Tech. rep. The Johns Hopkins University Press, published for The World Bank, 1997.

[23] Scrinis, Gyorgy and Monteiro, Carlos Augusto. “Ultra-processed foods and the limits of product reformulation”. In: *Public Health Nutrition* 21.1 (2018), pp. 247–252. DOI: 10.1017/S1368980017001392.

[24] Deaton, Angus. “Quality, Quantity, and Spatial Variation of Price”. In: *American Economic Review* 78 (3 1988), pp. 418–430.

[25] Maldonado, Norman, García, William, and Jimenez, Jairo. “Correcting bias on Estimation of Aggregate Consumption of Sweetened Beverages and potential substitutes in Colombia”. 2022.

[26] Herrán, Oscar F, Bermúdez, Jhael N, and Zea, María del Pilar. “Cambios alimentarios en Colombia; resultados de dos encuestas nacionales de nutrición, 2010-2015”. In: *Salud UIS* 52.1 (2019), pp. 21–31. DOI: 10.18273/revsal.v52n1-2020004.

[27] Caro, Juan Carlos et al. “Sugary drinks taxation, projected consumption and fiscal revenues in Colombia: Evidence from a QUAIDS model”. In: *PLOS ONE* 12.12 (2017), pp. 1–16. DOI: 10.1371/journal.pone.0189026.

[28] UN. *Classification of Individual Consumption According to Purpose (COICOP) 2018*. Tech. rep. ST/ESA/STAT/SER.M/99. Department of Economic and Social Affairs, Statistics Division, United Nations (UN), 2018.

[29] Babu, Suresh, Gajanan, Shailendra, and Hallam, J.Arne. *Nutrition Economics. Principles and Policy Applications*. Academic Press-Elsevier, 2017. ISBN: 978-0-12-800878-2. DOI: 10.1016/B978-0-12-800878-2.00018-9.

[30] Pollak, Robert and Wales, Terence. *Demand System Specification and Estimation*. Oxford University Press, 1992. ISBN: 0-19-506941-2.

[31] Abdelkrim, Araar et al. *Introducing WELCOM: A tool to simulate the welfare impacts of competition. Version 2.1*. World bank, 2018.

[32] Quirmbach, Diana et al. "Effect of increasing the price of sugar-sweetened beverages on alcoholic beverage purchases: an economic analysis of sales data". In: *Journal of Epidemiology and Community Health* 72 (4 Apr. 2018), p. 324. DOI: 10.1136/JECH-2017-209791.

[33] WCO. *The Harmonized System: A universal language for international trade 30 years on*. Tech. rep. World Customs Organization (WCO), 2018.

[34] UN. *Central Product Classification (CPC), Version 2.1*. Tech. rep. Department of Economic and Social Affairs, Statistics Division, United Nations (UN), 2015.

[35] Vecino-Ortiz, Andres I and Arroyo-Ariza, Daniel. "A tax on sugar sweetened beverages in Colombia: Estimating the impact on overweight and obesity prevalence across socio economic levels". In: *Social Science {\&} Medicine* 209 (2018), pp. 111–116. DOI: 10.1016/J.SOCSCIMED.2018.05.043.

[36] Babor, Thomas et al. *Alcohol: No Ordinary Commodity. Research and Public Policy*. 2nd.Ed. Oxford University Press, Pan American Health Organization, 2010. ISBN: 978-0-19-955114-9.

[37] WHO. *Global status report on alcohol and health 2018*. Geneva: World Health Organization (WHO), 2018. ISBN: 978-92-4-156563-9.

[38] PAHO. *Sugar-sweetened beverage taxation in the Region of the Americas*. Pan American Health Organization (PAHO), 2020. ISBN: 978-92-75-12300-3.

[39] Hattersley, Libby et al. *Countering common arguments against taxes on sugary drinks*. Tech. rep. Knowledge Brief. Health, Nutrition and Population Global Practice, World Bank Group, 2020.

A Summary statistics

Table 1: Household demographics

Variable	Mean	Std. dev.	Min	Max
HH Sex	0.63	0.48	0	1
HH Age	48.08	15.85	11	107
Rural	0.22	0.41	0	1
No members	3.37	1.73	1	22
Dummy solo member	0.14	0.35	0	1
Dummy children	0.27	0.44	0	1
HH Educ none	0.06	0.24	0	1
HH Educ primary	0.33	0.47	0	1
HH Educ Secundary	0.40	0.49	0	1
HH Educ higher	0.21	0.41	0	1
Total income	1,814,759	1,903,735	1,000	12,900,000
Income per member	668,827	890,345	200	12,800,000

Notes: N= 82,934

B Estimates of elasticities

Table 2: Beverage consumption statistics

Codified					
Off trade	**Proportion (2)**	**Expenditure (>0) (3)**	**Share (4)**	**Unit value (5)**	**Consumption (6)**
SB	0.65	4.38	0.24	1.00	3.18
Beer	0.07	11.67	0.03	1.18	0.67
Spirits	0.10	24.17	0.07	14.41	0.09
Water	0.17	3.39	0.04	0.32	3.25
Dairy	0.75	8.36	0.42	0.78	7.37
Full market					
SB	0.74	5.32	0.26	1.08	3.88
Beer	0.07	11.67	0.03	1.18	0.67
Spirits	0.15	23.06	0.08	7.09	0.30
Water	0.17	3.39	0.03	0.32	3.25
Dairy	0.75	8.42	0.38	0.78	7.50
Corrected					
Off trade	**Proportion**	**Expenditure (>0)**	**Share**	**Unit value**	**Consumption**
SB	0.67	4.00	0.26	0.72	3.99
Beer	0.07	10.44	0.03	1.38	0.10
Spirits	0.11	19.99	0.07	13.67	2.79
Water	0.18	2.90	0.05	0.32	0.48
Dairy	0.80	9.25	0.59	1.23	6.89
Full market					
SB	0.78	8.65	0.33	1.02	8.40
Beer	0.07	10.44	0.02	1.38	0.48
Spirits	0.16	19.12	0.07	7.39	0.27
Water	0.18	2.97	0.03	0.35	2.58
Dairy	0.86	12.07	0.55	1.49	8.80

Notes: (2) Proportion of households with positive consumption. (3) Average monthly expenditure of households with positive consumption (in US Dollars). (4) Share of good g y the total expenditure on the goods included in the basket. (5) Average unit value per liter (in US Dollars). (6) average monthly consumption in liters.

Table 3: Price elasticities and bias analysis

Variable	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Off trade						
SB	-0.355**	-0.348**	-0.408**	-0.367**	-0.355**	-0.768**
	[0.004]	[0.004]	[0.011]	[0.004]	[0.004]	[0.009]
Beer	-0.568**	-0.59**	-0.448**	-0.59**	-0.568**	-1.211**
	[0.014]	[0.013]	[0.076]	[0.015]	[0.014]	[0.023]
Spirits	-0.62**	-0.709**	-0.288**	-0.573**	-0.62**	-0.449**
	[0.011]	[0.012]	[0.047]	[0.011]	[0.011]	[0.015]
Water	-1.492**	-1.521**	-1.49**	-1.568**	-1.492**	-1.997**
	[0.005]	[0.004]	[0.011]	[0.006]	[0.005]	[0.007]
Diary	-0.185*	-0.19*	-0.158*	-0.19*	-0.185*	-0.151*
	[0.001]	[0.001]	[0.001]	[0.0001]	[0.0001]	[0.0009]
Full market						
SB	-0.503**	-0.502**	-0.492**	-0.529**	-0.356**	-0.562**
	[0.003]	[0.003]	[0.009]	[0.003]	[0.003]	[0.005]
Beer	-0.174**	-0.155**	1.441**	-0.12**	-0.31**	-1.299**
	[0.017]	[0.014]	[0.091]	[0.015]	[0.015]	[0.025]
Spirits	-0.767**	-0.807**	-0.557**	-0.786**	-0.78**	-0.703**
	[0.007]	[0.007]	[0.025]	[0.007]	[0.005]	[0.006]
Water	-1.473**	-1.528**	-1.563**	-1.564**	-1.518**	-1.857**
	[0.005]	[0.005]	[0.015]	[0.007]	[0.005]	[0.007]
Diary	-0.286*	-0.294*	-0.27*	-0.284*	-0.194*	-0.331*
	[0.001]	[0.001]	[0.002]	[0.0001]	[0.0001]	[0.0009]

Notes: Bootstrapped standard errors in brackets ** Statistical significance 1% * Statistical significance 5%. Model 1 base estimates. Model 2 without censoring correction. Model 3 without sampling weights. Model 4 without regional effects. Model 5 without consumption correction. Model 6 without rural households. All models include the full set of demographics controls.

Table 4: Price elasticity by household income tercile

Variable		**Tercile**		
Off trade	**Total**	**1**	**2**	**3**
SB	-0.355**	-0.364**	-0.323**	-0.247**
	[0.004]	[0.003]	[0.005]	[0.008]
Beer	-0.568**	-0.733**	-0.407**	-0.107**
	[0.014]	[0.02]	[0.03]	[0.026]
Spirits	-0.62**	-0.26**	-0.502**	-0.671**
	[0.011]	[0.019]	[0.022]	[0.011]
Water	-1.492**	-1.595**	-1.46**	-1.232**
	[0.005]	[0.016]	[0.012]	[0.012]
Dairy	-0.185*	-0.288**	-0.135**	-0.132**
	[0.001]	[0.001]	[0.001]	[0.001]
Full Market	**Total**	**1**	**2**	**3**
SB	-0.503**	-0.522**	-0.442**	-0.428**
	[0.003]	[0.004]	[0.004]	[0.006]
Beer	-0.174**	-0.429**	-0.163**	0.125**
	[0.017]	[0.02]	[0.038]	[0.039]
Spirits	-0.767**	-0.606**	-0.628**	-0.815**
	[0.007]	[0.009]	[0.01]	[0.007]
Water	-1.473**	-1.653**	-1.473**	-1.236**
	[0.005]	[0.015]	[0.012]	[0.01]
Dairy	-0.286*	-0.397**	-0.166**	-0.248**
	[0.001]	[0.002]	[0.001]	[0.001]

Notes: Bootstrapped standard errors in brackets ** Statistical significance 1% * Statistical significance 5%.

Table 5: Cross price elasticities

Variable	**SB**	**Beer**	**Spirits**	**Water**	**Dairy**
Off trade					
SB	-0.355**	0.029	0.034**	-0.003	-0.05**
	[0.004]	[0.03]	[0.011]	[0.019]	[0.001]
Beer	-0.133**	-0.568**	-0.143**	-0.431**	-1.223**
	[0.002]	[0.014]	[0.006]	[0.005]	[0.0001]
Spirits	-0.622**	-0.088**	-0.62**	-0.013	-1.8**
	[0.004]	[0.03]	[0.011]	[0.019]	[0.001]
Water	-0.328**	-0.232**	0.155**	-1.492**	0.153**
	[0.002]	[0.014]	[0.006]	[0.005]	[0.0001]
Dairy	-0.011*	0.002	0.003	0.013	-0.185*
	[0.001]	[0.014]	[0.01]	[0.008]	[0.001]
Variable	**SB**	**Beer**	**Spirits**	**Water**	**Dairy**
Full					
SB	-0.503**	-0.014	0.027**	0.004	-0.014**
	[0.003]	[0.034]	[0.007]	[0.019]	[0.001]
Beer	-1.084**	-0.174**	-0.162**	-0.268**	-0.558**
	[0.001]	[0.017]	[0.003]	[0.005]	[0.0001]
Spirits	-0.556**	-0.054	-0.767**	-0.015	-1.468**
	[0.003]	[0.034]	[0.007]	[0.019]	[0.001]
Water	-0.581**	-0.187**	0	-1.473**	-0.217**
	[0.001]	[0.017]	[0.003]	[0.005]	[0.0001]
Dairy	-0.001	0.009	-0.009	0.016*	-0.286*
	[0.001]	[0.01]	[0.006]	[0.007]	[0.001]

Notes: Bootstrapped standard errors in brackets ** Statistical significance 1% * Statistical significance 5%.

Table 6: Income elasticities

Variable	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Off trade						
SB	0.88**	0.9**	0.7**	0.88**	0.88**	0.59**
	[0.0315]	[0.0251]	[0.1081]	[0.0354]	[0.0284]	[0.0602]
Beer	3.52**	1.24**	1.78**	3.52**	3.52**	4.91**
	[0.0315]	[0.0251]	[0.1081]	[0.0354]	[0.0284]	[0.0602]
Spirits	5.31**	1.18**	4.88**	4.31**	4.31**	3.54**
	[0.0103]	[0.0108]	[0.0449]	[0.0115]	[0.0115]	[0.0144]
Water	2.04**	1.09**	1.35**	2.04**	2.04**	1.12**
	[0.019]	[0.0163]	[0.026]	[0.0187]	[0.0191]	[0.0161]
Diary	0.89**	0.86**	0.79**	0.89**	0.89**	0.96**
	[0.0028]	[0.0026]	[0.0041]	[0.0027]	[0.0029]	[0.0054]
Full market						
SB	0.88**	0.87**	0.81**	0.88**	0.9**	0.74**
	[0.0053]	[0.0062]	[0.013]	[0.006]	[0.0072]	[0.008]
Beer	3.32**	1.25**	2.39**	3.32**	3.4**	5.22**
	[0.0317]	[0.0375]	[0.1563]	[0.0359]	[0.0323]	[0.08]
Spirits	3.22**	1.12**	2.96**	3.22**	2.88**	2.92**
	[0.007]	[0.007]	[0.0336]	[0.0073]	[0.0055]	[0.0076]
Water	3.08**	1.12**	1.86**	3.08**	2.32**	2.28**
	[0.0217]	[0.0192]	[0.0356]	[0.019]	[0.0178]	[0.0198]
Diary	0.86**	0.82**	0.84**	0.86**	0.91**	0.73**
	[0.0029]	[0.0033]	[0.0057]	[0.003]	[0.0028]	[0.0053]

Notes: Bootstrapped standard errors in brackets ** Statistical significance 1% * Statistical significance 5%. Model 1 base estimates. Model 2 without censoring correction. Model 3 without sampling weights. Model 4 without regional effects. Model 5 without consumption correction. Model 6 without rural households. All models include the full set of demographics controls.

Dear Editor of Food Policy:

We are pleased to submit an original research article entitled "Revisiting Elasticities of Sweetened Beverages in Colombia", by Norman Maldonado, William García, and Elisa Cadena. The problem addressed by this paper is the identification of biases in estimates of price and income elasticities of Sweetened Beverages and the methods to correct those biases to obtain more precise estimates of elasticities. Such precision is crucial for an accurate design of health taxes and ex-ante evaluation of the effects of such policy.

The empirical approach we follow takes advantage of a nationally representative survey, regular expression analytics, statistical inference, demand systems and microeconometrics to accurately measure consumption of SB and potential substitutes and to remove the biases from estimates of elasticities. The current work on the area estimates elasticities using data only on coded consumption and with adjustment of some, but not all, the sources of bias identified in this paper; results from that work suggest elasticities lower than one, and for Colombia, the only estimation available is -1.6. We claim that those estimates have a systematic set of downward biases that, when adjusted, produce estimates close to -0.5; such difference has enormous implications on the design of taxes on Sweetened Beverages.

We submit the paper to this journal because unhealthy diets have been identified as one of the top causes of mortality worldwide, taxes on SB have been identified as a potentially effective fiscal policy for the primary prevention of such mortality, and unbiased estimates of such elasticities are crucial for accurate design of taxes on Sugar-Sweetened Beverages. Thus, the evidence provided in the paper is expected to provide a method to reduce biases in the estimation of elasticities of Sweetened beverages using household surveys and, therefore, to improve the scientific quality of the discussion on taxes on Sweetened Beverages. In addition, the application of the method for the case of Colombia produced results that are relevant for the policy discussion in the country, making both the method and the results relevant for decision-making at the country level.

Look forward to your favorable consideration. Regards,

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Highlights

- We estimate a Unit Value Demand System Model for Sweetened Beverages in Colombia.
- We derive price and expenditure elasticities for households considering on-trade (away from home) or on-trade plus off-trade (Full market) consumption as a reaction to price or income change.
- Biases in elasticities can considerably mislead tax policy discussions. We document that censoring, exclusion of rural households, and omission of uncoded consumption are important sources of bias.